Huawei Access Node WHI-ACC-50 Hardware/Environmental Fault Resolution SOP

Node: WHI-ACC-50 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_7047

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for hardware/environmental faults on the Huawei Access Node WHI-ACC-50 located at the Whitefield East POP. The failure domain in question pertains to ASIC memory parity errors and line card crashes, precipitated by the loss of power supply redundancy. This scenario is characteristic of the Access role and Huawei equipment, which employ a Layer 2 switching architecture with a focus on high-speed packet forwarding. The underlying mechanics of the failure domain are rooted in the asynchronous transfer mode (ATM) cell switching and the use of Application-Specific Integrated Circuits (ASICs) to manage cell processing and queuing. In the event of a hardware fault, such as a line card crash or power supply redundancy loss, the node may experience a loss of packet forwarding capability, leading to a network outage. The Huawei CLI commands employed in this SOP are specifically designed to aid in the detection and isolation of hardware faults, as well as the implementation of corrective measures to restore node functionality.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Whitefield East POP to assist with physical access and equipment inspection.

- Verify the status of the redundant link to ensure it is operational and not contributing to the failure domain.

- Check the power supply redundancy status to confirm the loss of redundancy has occurred.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display board information, Expected Output: Board ID: WHI-ACC-50-1-1, Description: This command is used to display the board information, including the board ID, slot number, and firmware version. The expected output confirms the board in question is the WHI-ACC-50-1-1, which is the line card responsible for packet forwarding.
```

```
Command: display power supply information, Expected Output: Power supply redundancy: Lost, Description: This command is used to display the power supply information, including the redundancy status. The expected output confirms the power supply redundancy has been lost, contributing to the failure domain.
```

Resolution Steps

Step 1: Step: Inspect the line card for physical damage or signs of malfunction, Command: inspect board WHI-ACC-50-1-1, Description: This command is used to inspect the line card for physical damage or signs of malfunction. The output will indicate if any issues are present.

Step 2: Step: Replace the failed line card, Command: replace board WHI-ACC-50-1-1, Description: This command is used to replace the failed line card with a new one. The new line card will be configured with the same settings as the original.

Step 3: Step: Verify the power supply redundancy status, Command: display power supply information, Description: This command is used to verify the power supply redundancy status. The output will confirm if the power supply redundancy has been restored.

Post-Check Verification

```
Command: display board information, Expected Output: Board ID: WHI-ACC-50-1-1, Description: This command is used to verify the board information, including the board ID, slot number, and firmware version. The expected output confirms the board in question is the WHI-ACC-50-1-1, which is the line card responsible for packet forwarding.

Command: display power supply information, Expected Output: Power supply redundancy: Normal, Description: This command is used to verify the power supply redundancy status. The expected output confirms the power supply redundancy has been restored.
```

Roll-back Procedures

- Step: Restore the original line card, Command: restore board WHI-ACC-50-1-1, Description: This command is used to restore the original line card. The original line card will be reconfigured with the same settings as before.

- Step: Verify the power supply redundancy status, Command: display power supply information, Description: This command is used to verify the power supply redundancy status. The output will confirm if the power supply redundancy has been restored.